Treatment Assignment Design Evaluation for Spatial CRTs

Simulation Study with Heterogeneous Incidence — Results Summary

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1 Introduction

This report summarizes a simulation study evaluating **8 treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured. The central research question is:

> *Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?*

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000 (Mirzaei et al.), consistent with county-level analyses by Gan et al. and Watson et al.

The study uses Maximum Likelihood Estimation via a spatial autoregressive model (MLE) as the primary estimator, evaluated across 2,560 parameter scenarios.

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2 Mathematical Framework

2.1 Study Region and Spatial Structure

The simulation operates on a **regular lattice grid** $\mathcal{G} = \{s_1, \dots, s_N\}$ on $[1, G] \times [1, G]$, with $N = G^2$ clusters ($G = 10$, $N = 100$).

Two contiguity structures define neighbor relationships:

- **Rook contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge (up to 4 neighbors)
- **Queen contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge or corner (up to 8 neighbors)

The spatial weight matrix $\mathbf{W}$ has entries $w_{ij} = |\mathcal{N}(s_i)|^{-1}$ if $s_j \in \mathcal{N}(s_i)$ and 0 otherwise (row-standardized).

2.2 Outcome Data Generating Process

Outcomes follow a **Spatial Durbin Model (SDM)**:

$$\mathbf{Y} = (\mathbf{I} - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \gamma \cdot \mathrm{Spill}(\mathbf{Z}) + \beta\mathbf{X} + \varepsilon\right]$$

Parameter definitions:

Symbol	Role	Value / Range
τ	Direct treatment effect (estimand)	1.0 (fixed)
ρ	Outcome spatial autocorrelation	{0, 0.01, 0.2, 0.5}
γ	Spillover magnitude	{0.5, 0.6, 0.7, 0.8}
β	Incidence effect on outcome	1.0 (fixed)
σ	Residual standard deviation	1.0 (fixed)

Spillover construction:

$$\mathrm{Spill}_i(\mathbf{Z}) = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|}$$

Two spillover modes are evaluated:

- **control_only:** γ applied only to units with $Z_i = 0$
- **both:** γ applied to all units regardless of treatment status

Stationarity requires $|\rho| < 1/\lambda_{\max}(\mathbf{W})$; all swept ρ values satisfy this condition.

2.3 Incidence Generation

Three modes generate the cluster-level covariate $X_i \in [0, 1]$, each representing a different assumption about how historical incidence arises:

Mode 1 — iid Uniform:

$$X_i \overset{\text{iid}}{\sim} \text{Uniform}(0, 1)$$

No spatial structure; serves as the baseline. One config (rho_X = N/A).

Mode 2 — Spatial (SAR filter):

$$\mathbf{X}^* = (\mathbf{I} - \rho_X \mathbf{W})^{-1}\mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$X_i = \Phi(X_i^*) \in (0, 1)$$

Spatially correlated via SAR filter; probability integral transform (Φ) ensures X_i is marginally Uniform$(0, 1)$ but spatially dependent. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Mode 3 — Poisson (count rates):

1. Spatially correlated log-rates: $\log \lambda_i^* = \mu_0 + [(\mathbf{I} - \rho_X \mathbf{W})^{-1}\mathbf{u}]_i$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 1{,}000$ (equal population per cluster)
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5)/N$

Base rate $\mu_0 = \log(35/100{,}000)$ mimics SUD incidence. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence is generated **once per (mode, ρ_X) configuration** and held fixed across all (ρ, γ, nb_type) scenarios. This reflects the realistic constraint that a researcher observes historical incidence once before designing the trial; spatial model parameters affect the outcome DGP, not the incidence itself.

2.4 Treatment Assignment Designs

Table 2: 8 treatment assignment strategies evaluated in this simulation

ID	Name	Deterministic?
1	Design 1: Checkerboard	Yes
2	Design 2: High Incidence Focus	Yes
3	Design 3: Saturation Quadrants	No
4	Design 4: Isolation Buffer	No
5	Design 5: 2x2 Blocking	No
6	Design 6: Balanced Quartiles	No

ID	Name	Deterministic?
7	Design 7: Balanced Halves	No
8	Design 8: Incidence-Guided Saturation Quadrants	No

Designs 1 and 2 are **deterministic** given the incidence vector: they produce the same assignment on every resample. The simulation detects this and runs each once, replicating the result across all design resamples (saving ~25x compute).

2.5 Estimation Method

Maximum Likelihood (MLE via SAR):

Fits `lagsarlm(Y ~ Z + Spill + X, listw = W)` from the `spatialreg` package, simultaneously estimating ρ, τ, and regression coefficients. The true spillover covariate Spill($\mathbf{Z}$) is included as an **oracle regressor** (i.e., the researcher is assumed to know the true spillover structure). CIs are derived from the asymptotic covariance matrix.

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3 Simulation Algorithm

3.1 Parameter Grid

Table 3: Simulation parameter grid (2,560 total scenarios)

Parameter	Values	Levels
Incidence mode	iid, poisson, spatial	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	queen, rook	2
rho (outcome spatial autocorr.)	0, 0.01, 0.2, 0.5	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	both, control_only	2
design_id	1, 2, 3, 4, 5, 6, 7, 8	8
TOTAL		2,560

3.2 Resampling Structure

For each scenario, the simulation draws:

Mode	Design resamples (R_d)	Outcome resamples (R_o)	Iterations / scenario
MLE	25	10	250

3.3 Evaluation Metrics

For each scenario, across K total iterations (design $\times$ outcome resamples):

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{SD} = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(\hat{\tau}_k - \bar{\hat{\tau}})^2}$$

$$\text{MSE} = \text{Bias}^2 + \text{SD}^2$$

$$\text{Coverage} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{1}\left[\tau \in (\hat{\tau}_{k,L}, \hat{\tau}_{k,U})\right]$$

3.4 Per-Scenario Reproducibility

Each scenario uses a deterministic seed:

$$\text{seed}_s = \texttt{digest::digest2int}(\texttt{paste}(\text{inc_mode}, \rho_X, \text{nb_type}, \rho, \gamma, \text{spill_type}, d_{\text{id}}, \text{sep} = \text{"|"}))$$

This ensures results are reproducible and order-independent: adding or removing scenarios from the parameter sweep does not affect other scenarios' seeds.

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4 Results: Maximum Likelihood Estimation (MLE)

4.1 Overall Summary

Table 5: MLE: Summary statistics across all 2,560 scenarios

Metric	Min	Mean	Max
Bias	-0.898	-0.029	0.455
MSE	0.012	0.206	4.061
Coverage	0.000	0.890	1.000
Fail Rate	NA	0.000	NA

4.2 Design Comparison by MSE

Table 6: MLE: Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
Design 8	0.019	0.064	0.079	0.939	0
Design 3	0.022	0.065	0.080	0.937	0
Design 6	0.012	0.073	0.103	0.937	0
Design 7	0.014	0.082	0.108	0.937	0
Design 4	0.001	0.144	0.148	0.940	0
Design 2	0.016	0.090	0.156	0.937	0
Design 5	0.008	0.128	0.229	0.938	0
Design 1	-0.322	0.451	0.744	0.553	0

4.3 MSE by Design and Neighbor Type

4.4 MSE Heatmap

4.5 Design Rank Frequency

Table 7: MLE: Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank (Rank 1 = lowest MSE).

Design	Rank_2	Rank_3	Rank_4	Rank_6	Rank_7	Rank_8	Rank_1	Rank_5
Design 3	20.3	17.2	14.1	1.6	0.0	0.0	34.4	12.5
Design 8	34.4	14.1	7.8	6.2	0.0	0.0	23.4	14.1
Design 6	14.1	14.1	26.6	7.8	0.0	0.0	18.8	18.8
Design 2	9.4	7.8	3.1	39.1	14.1	3.1	17.2	6.2
Design 7	10.9	18.8	25.0	12.5	0.0	0.0	6.2	26.6
Design 1	7.8	4.7	1.6	7.8	9.4	68.8	0.0	0.0

Design	Rank_2	Rank_3	Rank_4	Rank_6	Rank_7	Rank_8	Rank_1	Rank_5
Design 4	1.6	17.2	18.8	3.1	23.4	26.6	0.0	9.4
Design 5	1.6	6.2	3.1	21.9	53.1	1.6	0.0	12.5

4.6 Robustness Profile

Table 8: MLE: Robustness profile for iid Uniform incidence. Sorted by Median MSE. IQR = interquartile range across scenarios.

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0351	0.0447	0.0646	0.1105	0.1544	0.0658
Design 8	0.0363	0.0454	0.0677	0.1114	0.1753	0.0661
Design 6	0.0358	0.0421	0.0757	0.1576	0.2512	0.1155
Design 7	0.0327	0.0449	0.0827	0.1590	0.2818	0.1141
Design 2	0.0278	0.0530	0.1043	0.2042	0.3970	0.1512
Design 5	0.0426	0.0560	0.1195	0.3031	0.7737	0.2471
Design 4	0.1010	0.1281	0.1487	0.1659	0.1964	0.0377
Design 1	0.0356	0.1083	0.4729	0.9530	1.9159	0.8447

4.7 Coverage by Design

4.8 Bias-Variance Decomposition

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5 Sensitivity Analysis

5.1 Parameter Sensitivity

Table 9: MLE: Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence. Higher values indicate parameters where design choice matters more.

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0301	0.0123	0.4164	0.0954
Design 2	0.2274	0.0000	0.5639	0.0091
Design 3	0.0200	0.0018	0.8354	0.0342

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 4	0.1691	0.0123	0.0002	0.5471
Design 5	0.0045	0.0008	0.5781	0.2035
Design 6	0.0049	0.0038	0.7908	0.0808
Design 7	0.0014	0.0011	0.7864	0.0870
Design 8	0.0543	0.0029	0.8104	0.0242

5.2 Effect of Rho on Design Rankings

```
$`Design 1`
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```
$`Design 2`
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```
$`Design 3`
```

```
$`Design 4`
```

```
$`Design 5`
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```
$`Design 6`
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```
$`Design 7`
```

```
$`Design 8`
```

5.3 Incidence Mode Comparison

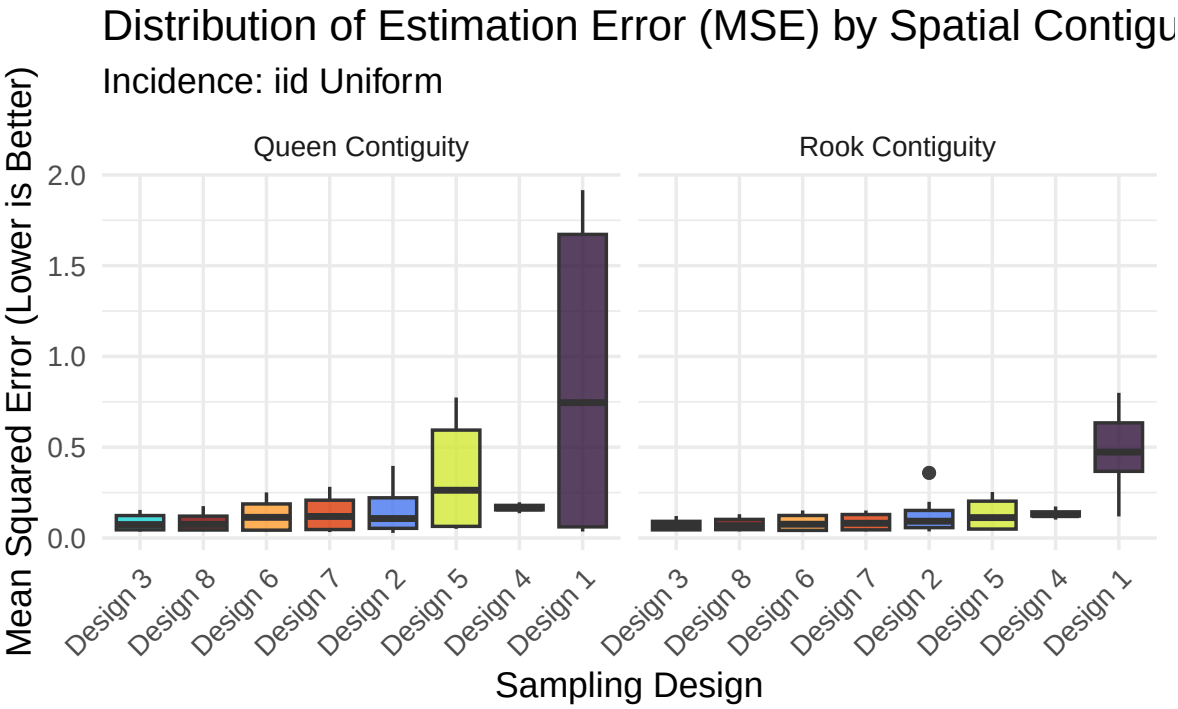

Figure 1: MLE: Distribution of MSE by design, faceted by spatial contiguity structure. Boxes sorted by median MSE within each panel.

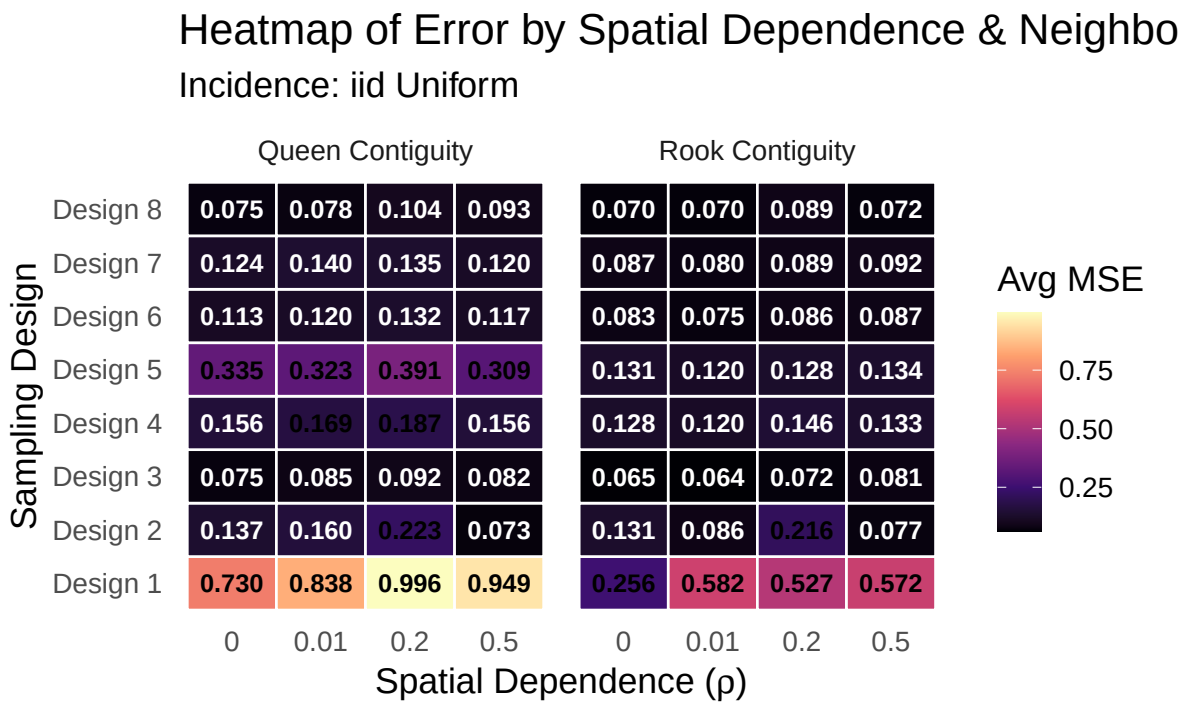

Figure 2: MLE MSE heatmap for iid Uniform incidence. Cells show average MSE across spillover type and gamma, faceted by neighbor type.

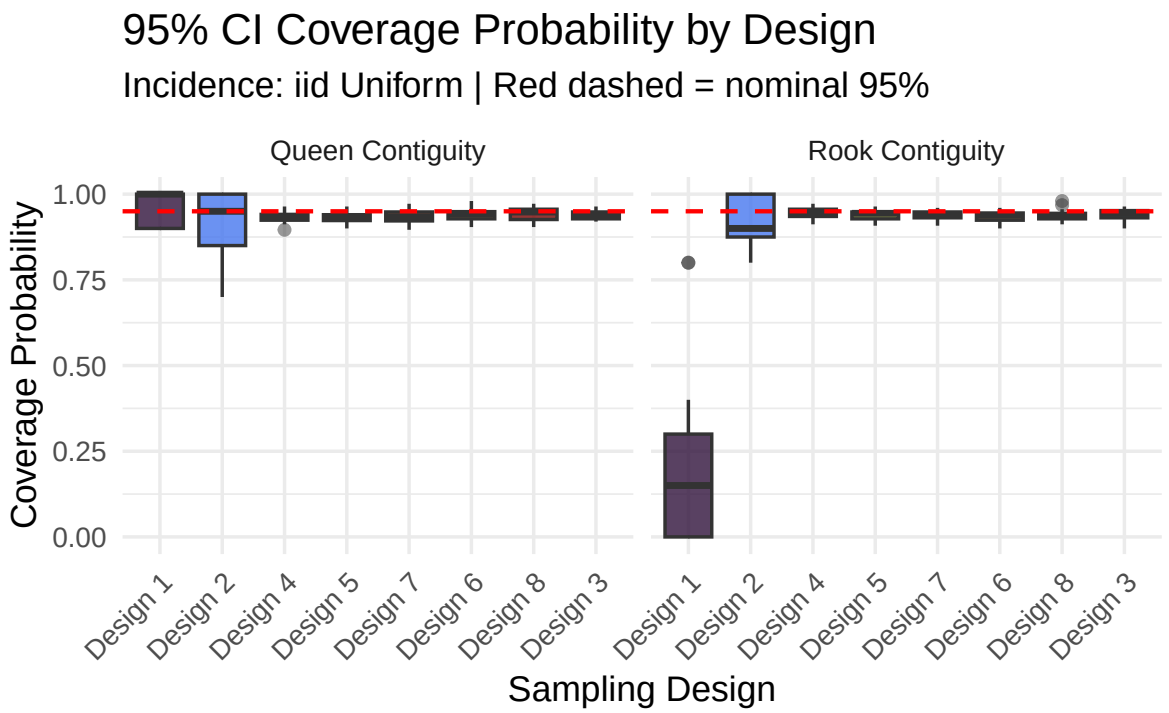

Figure 3: MLE: Coverage probability distribution by design across all scenarios. Red dashed line = nominal 95% target.

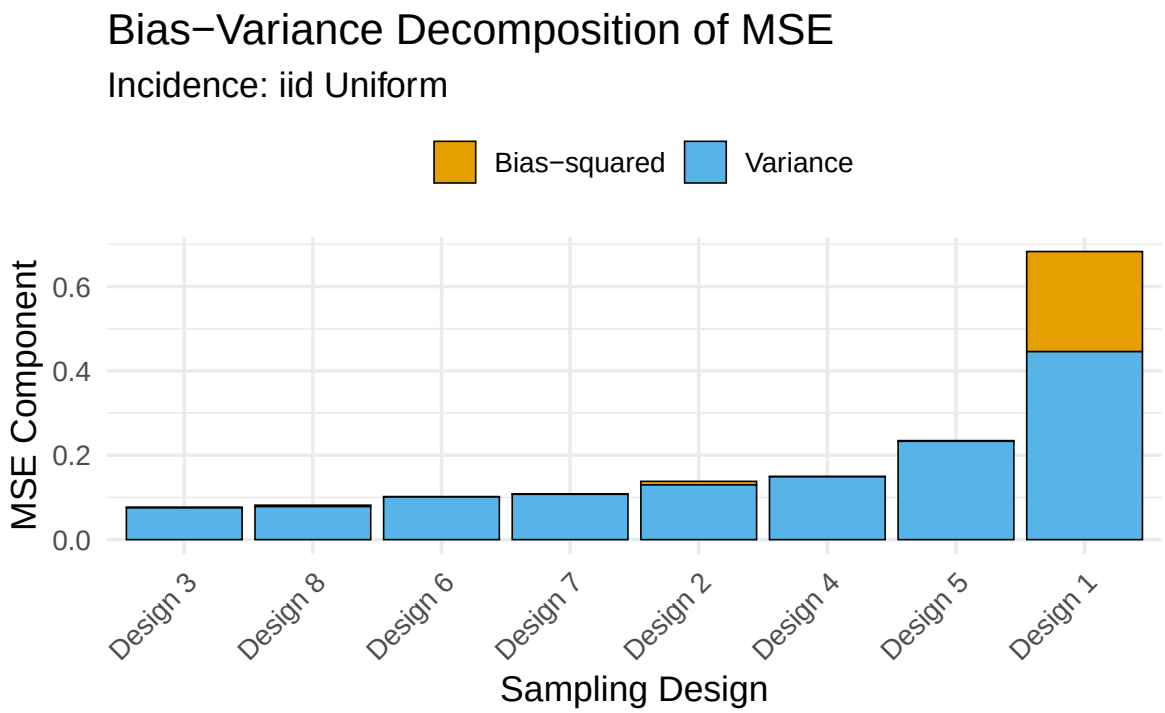

Figure 4: MLE: Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence. Total bar height = MSE.

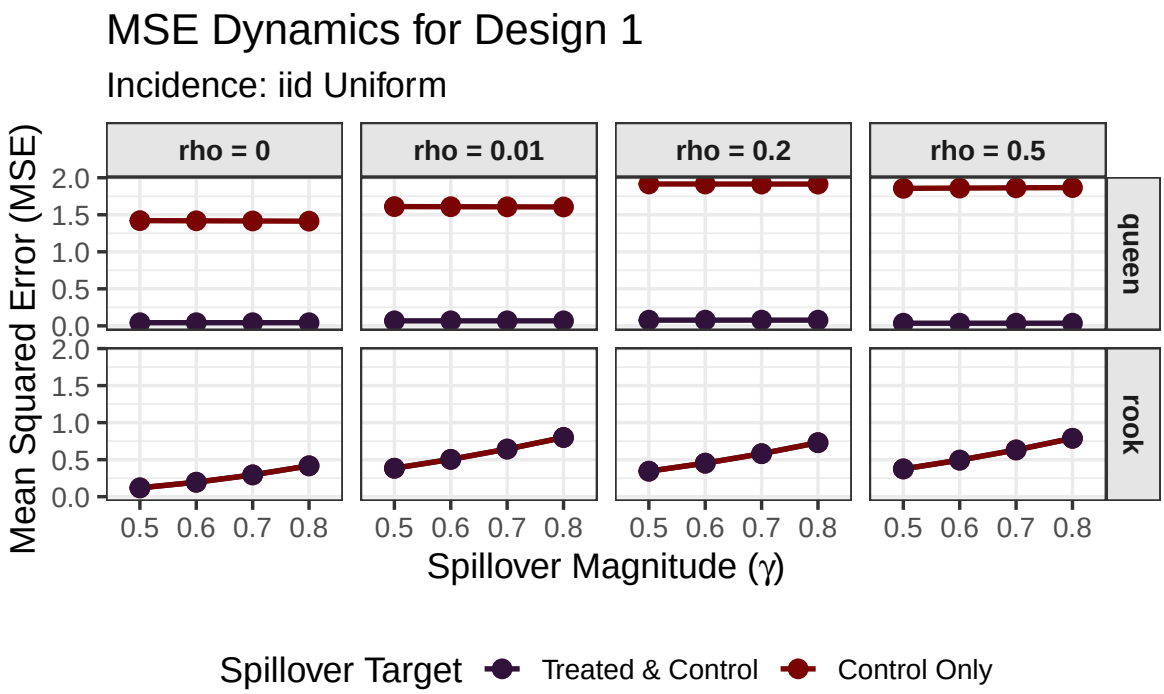

Figure 5: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

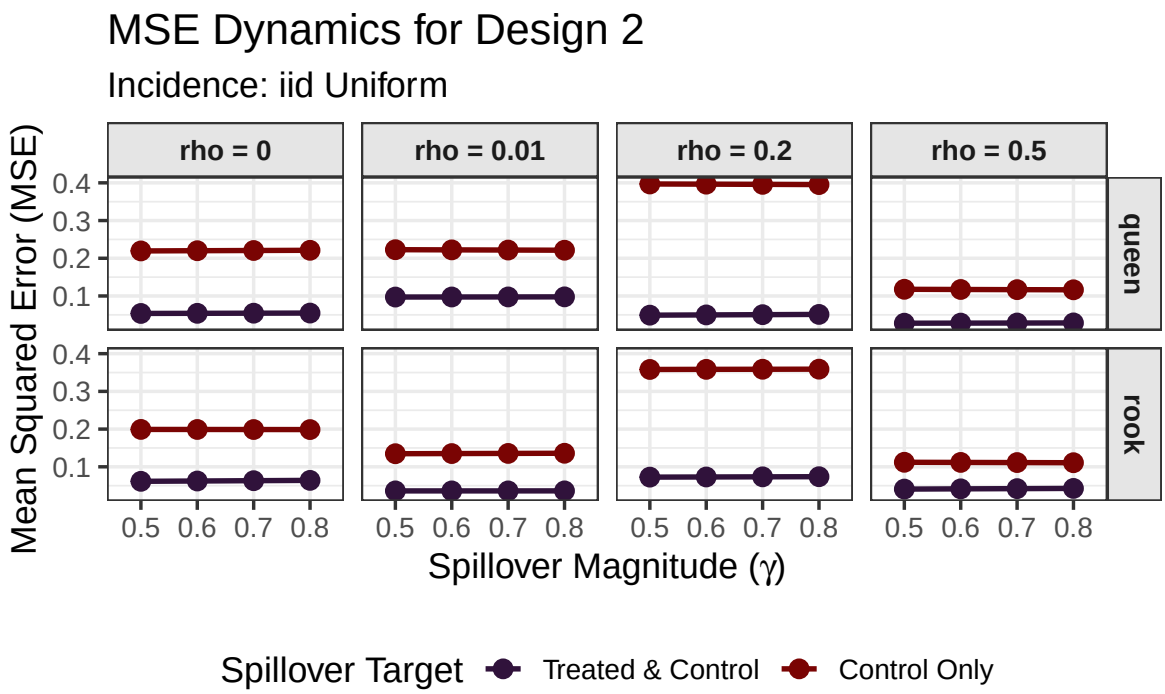

Figure 6: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

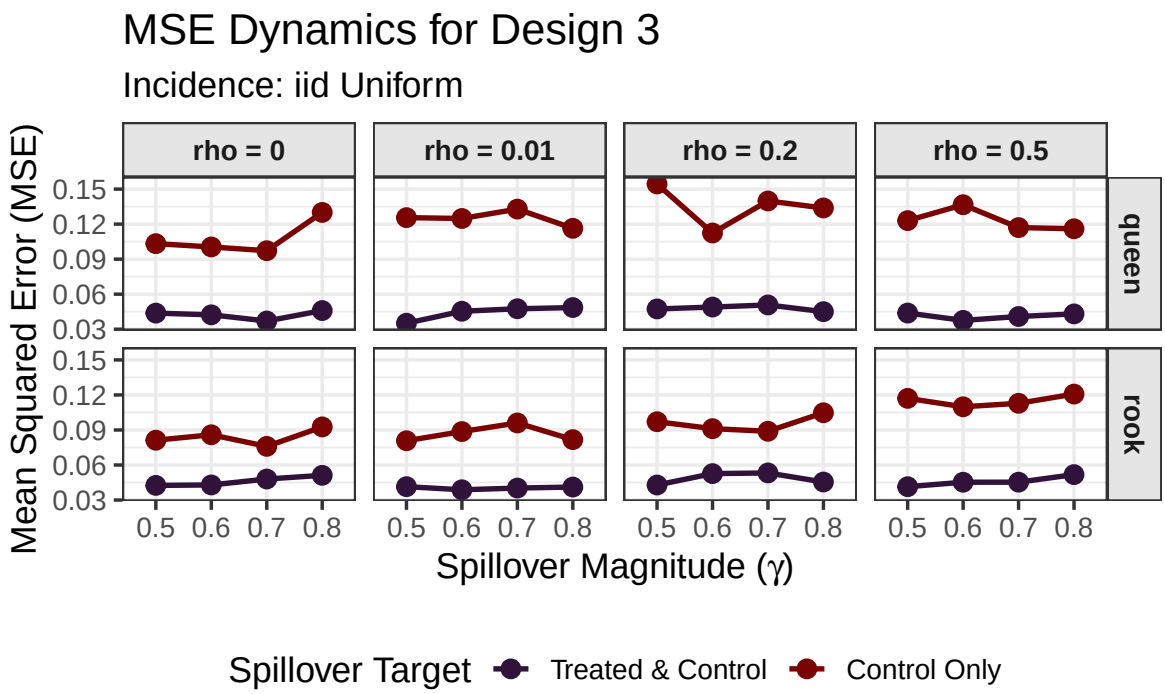

Figure 7: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

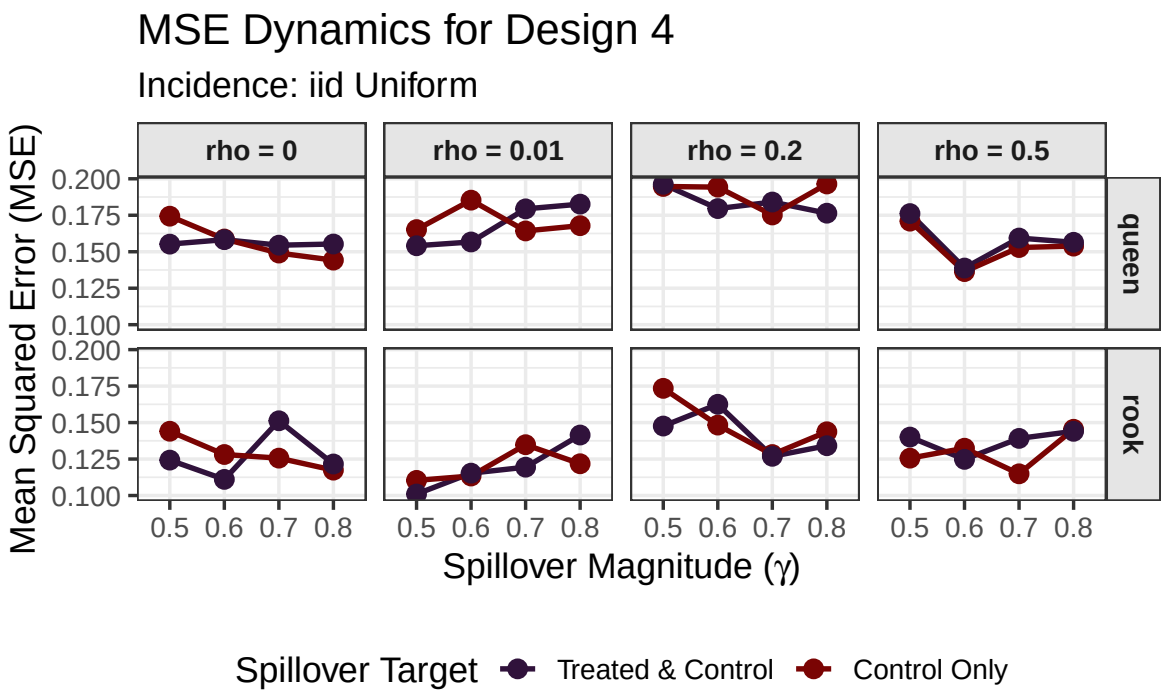

Figure 8: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

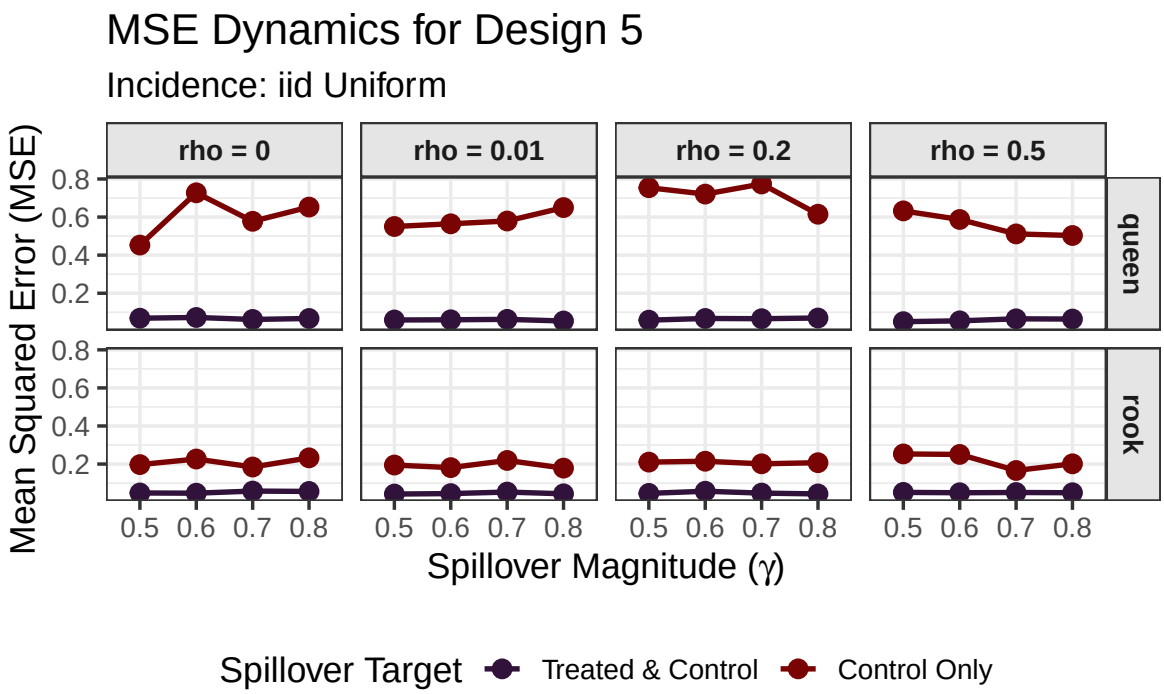

Figure 9: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

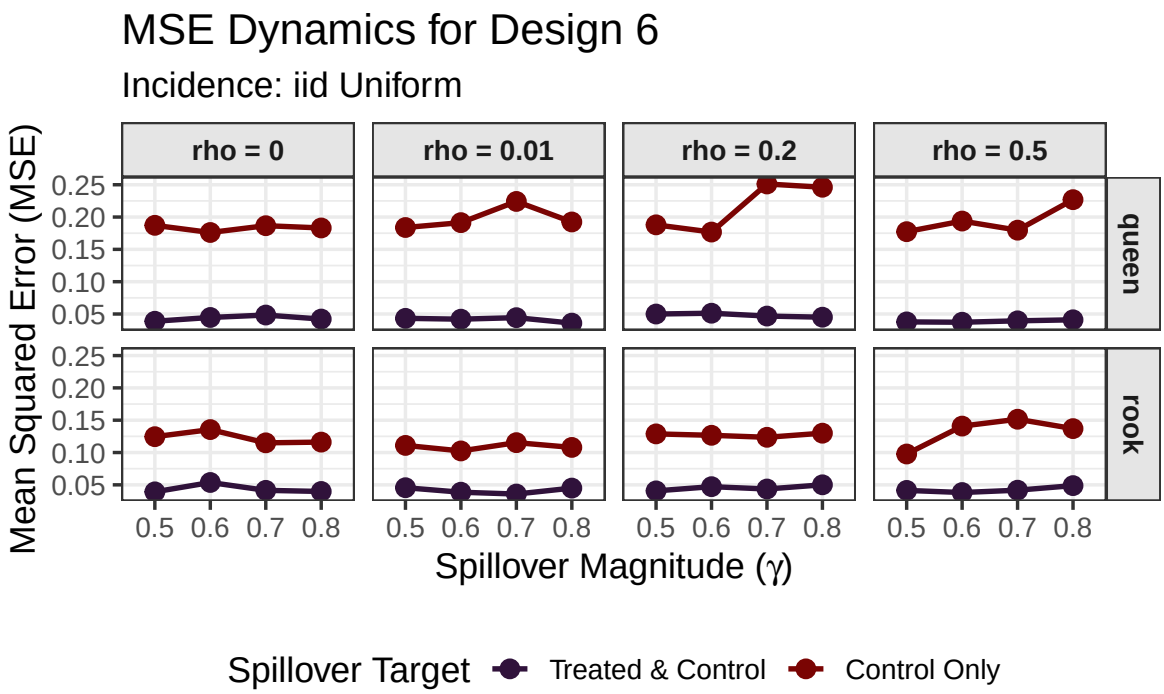

Figure 10: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

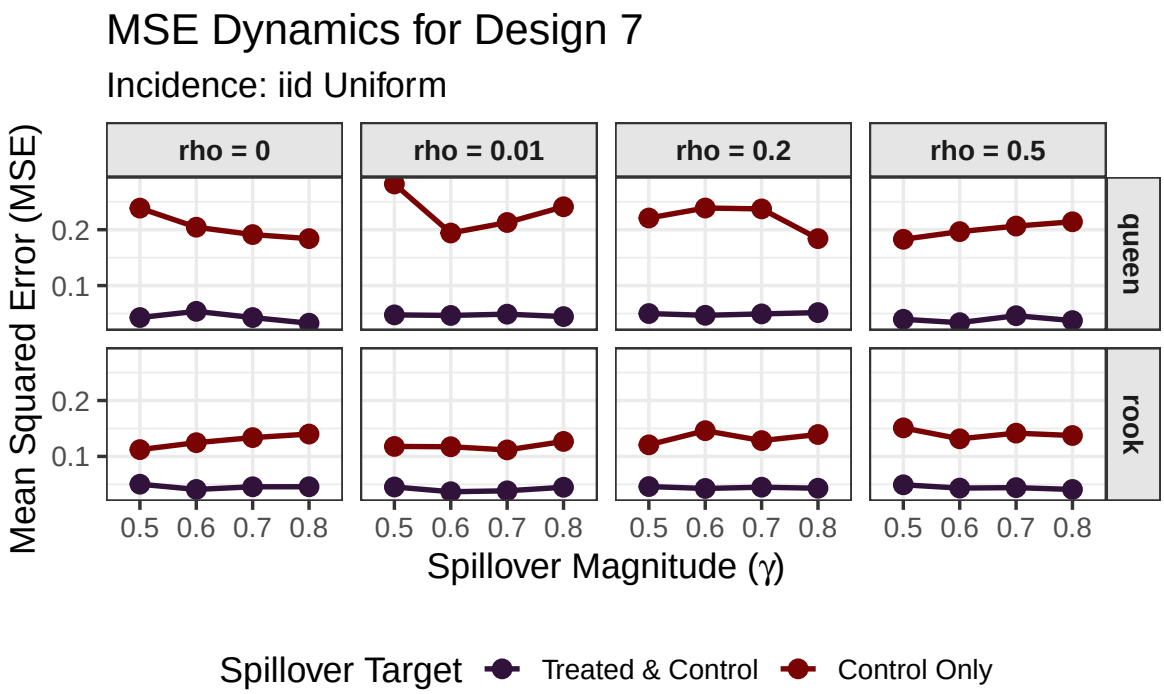

Figure 11: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

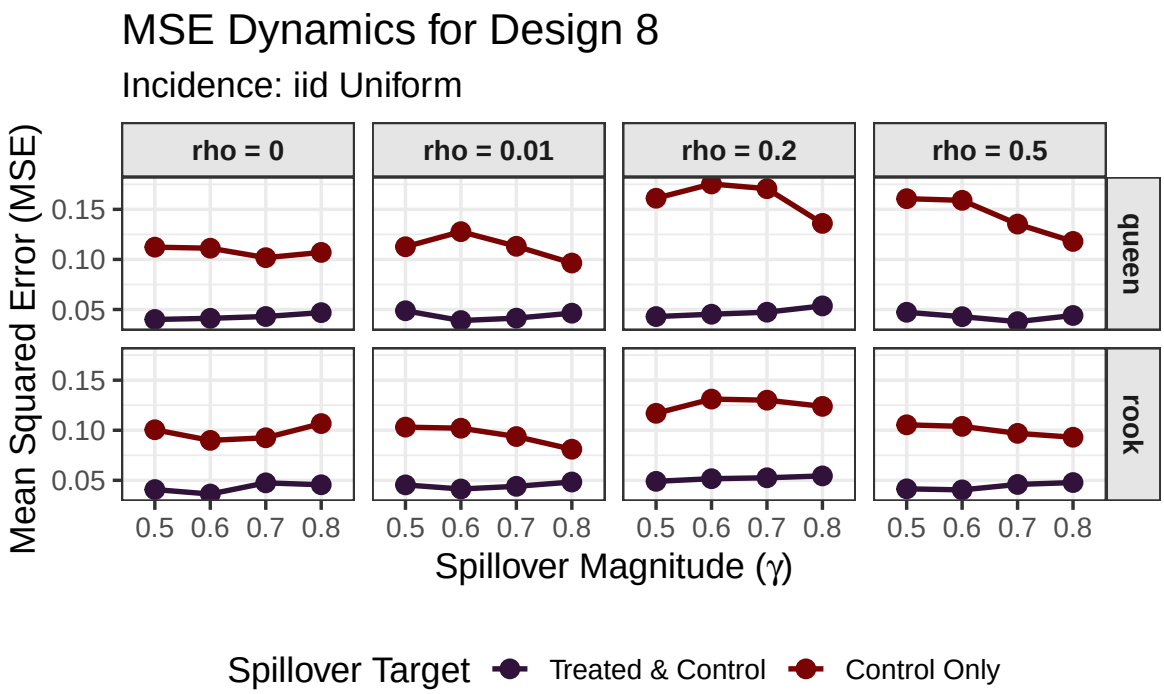

Figure 12: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

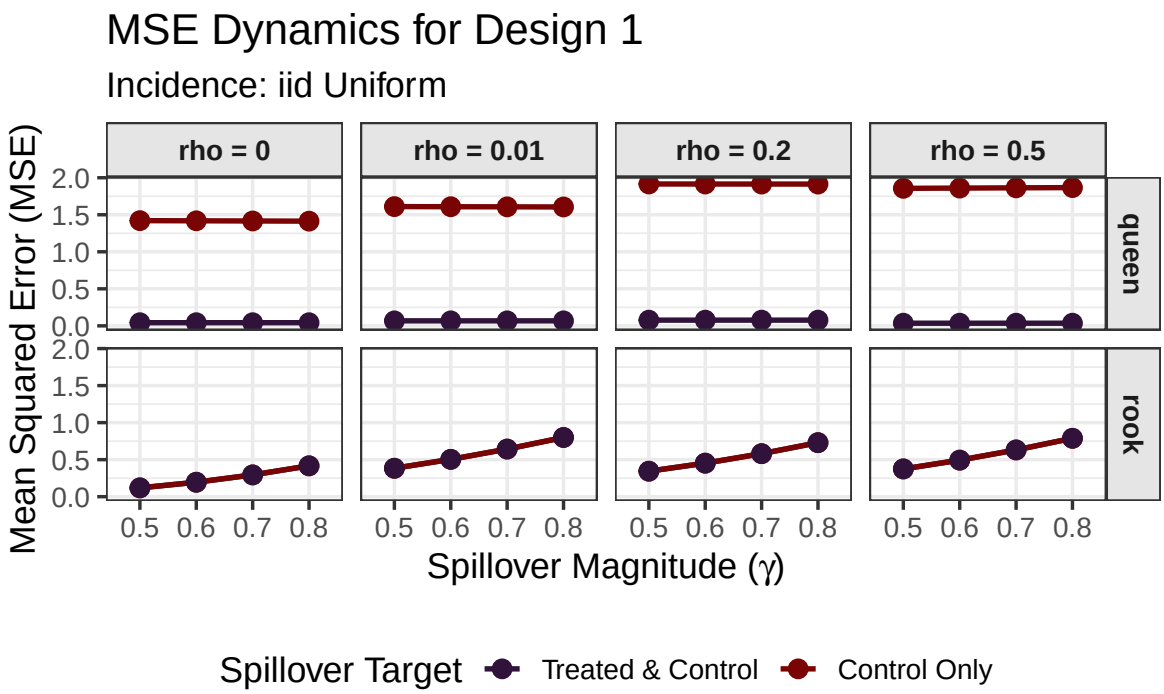

Figure 13: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

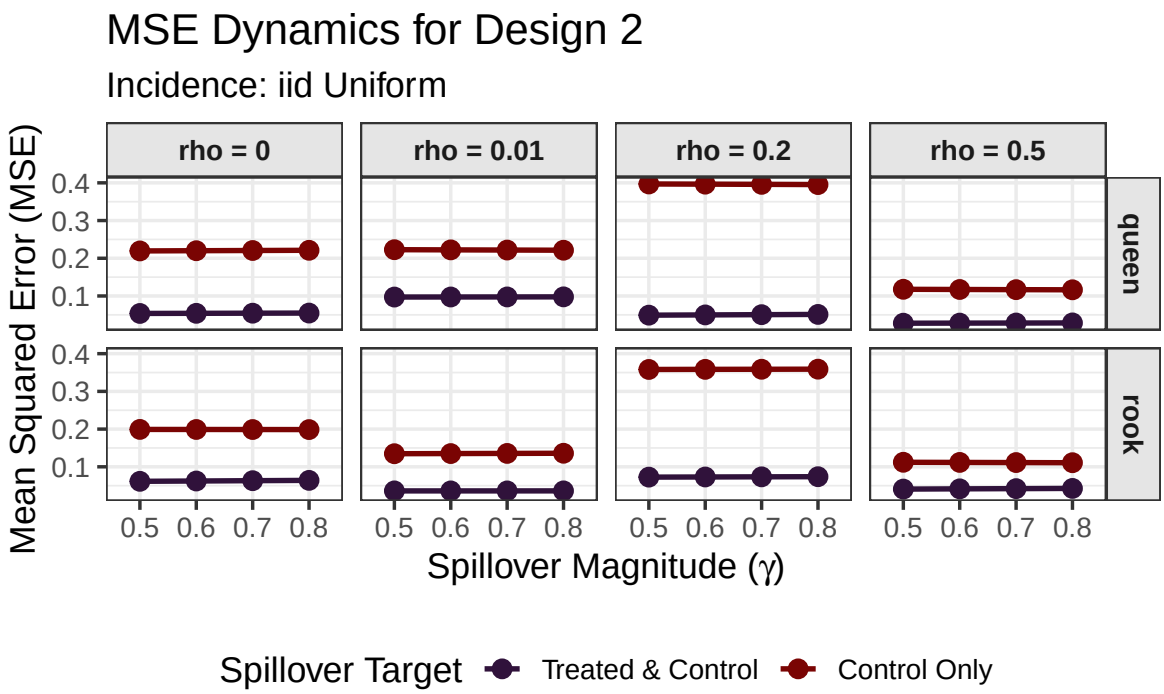

Figure 14: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

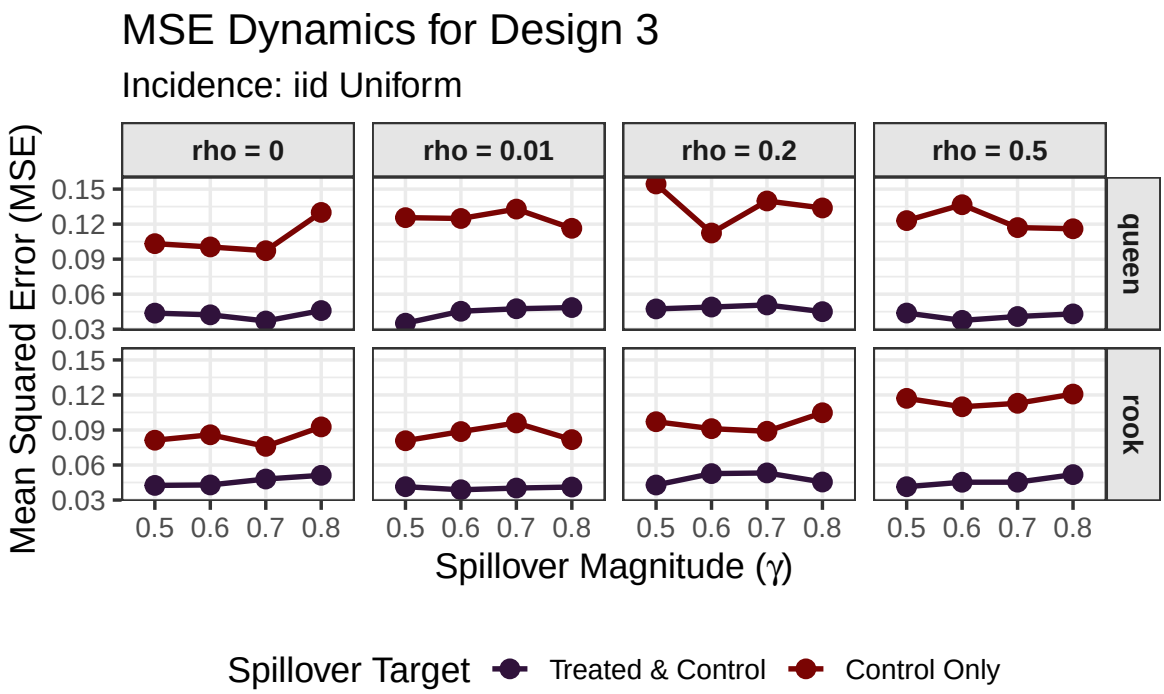

Figure 15: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

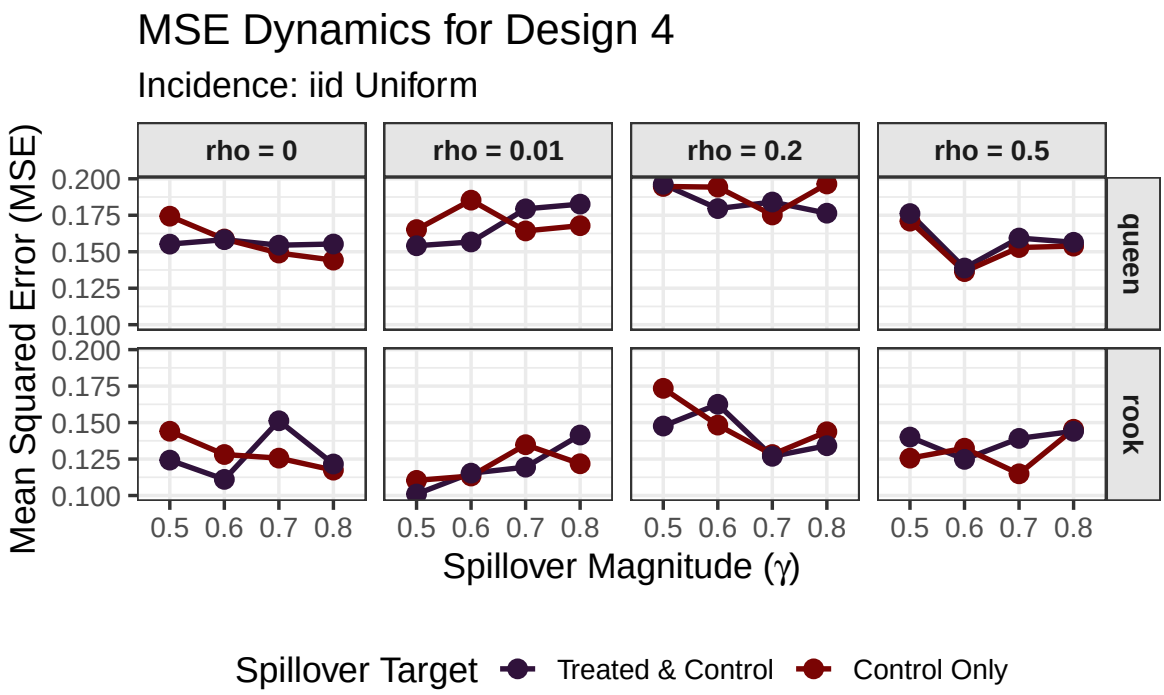

Figure 16: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

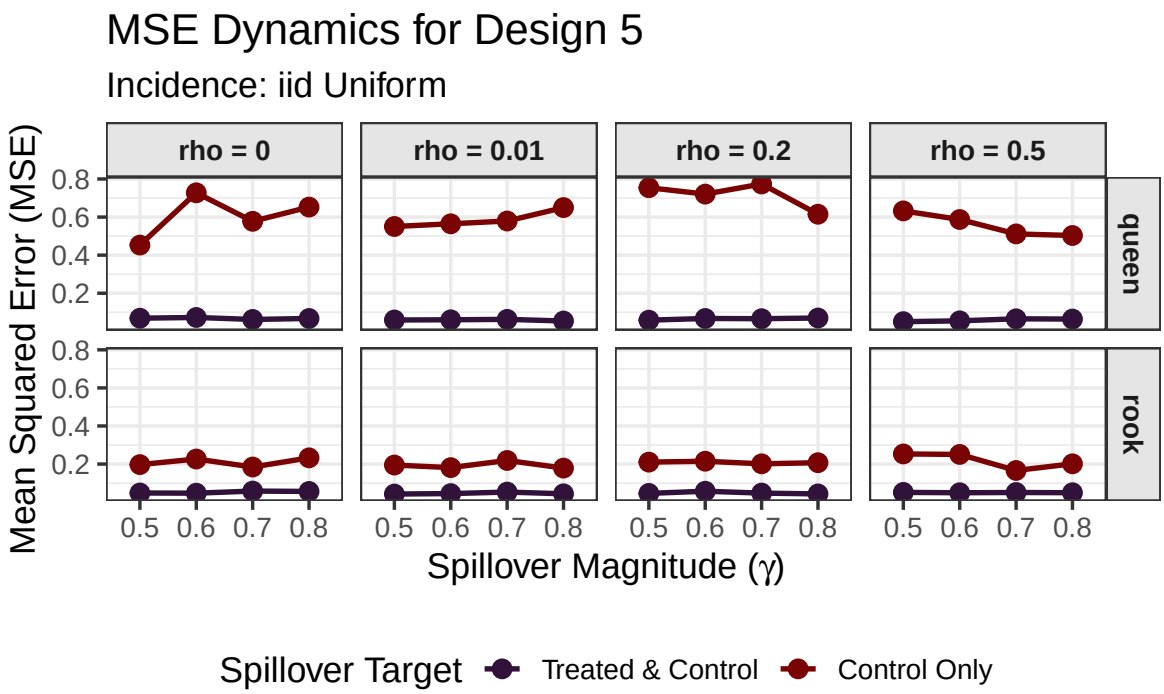

Figure 17: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

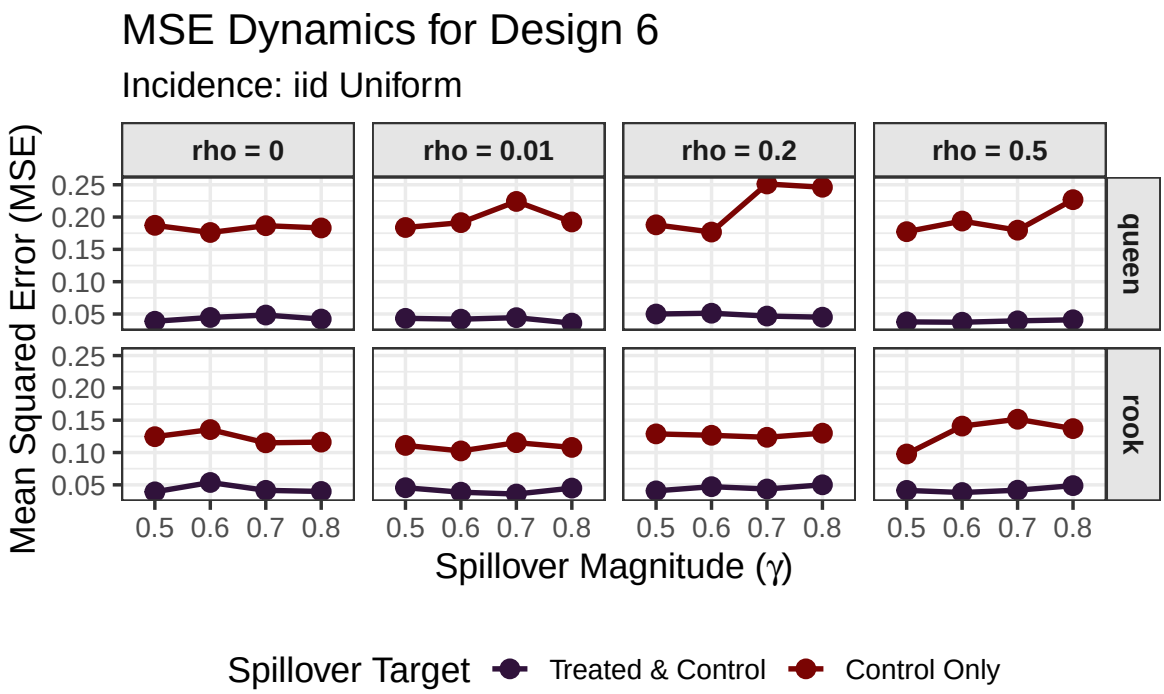

Figure 18: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

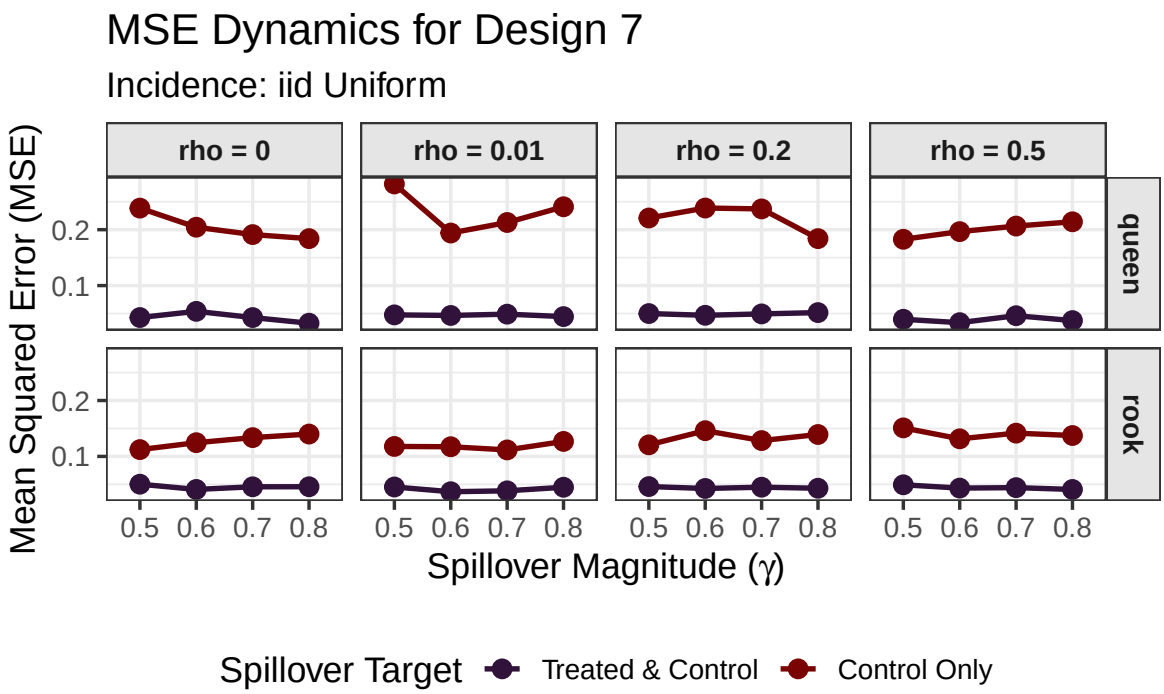

Figure 19: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

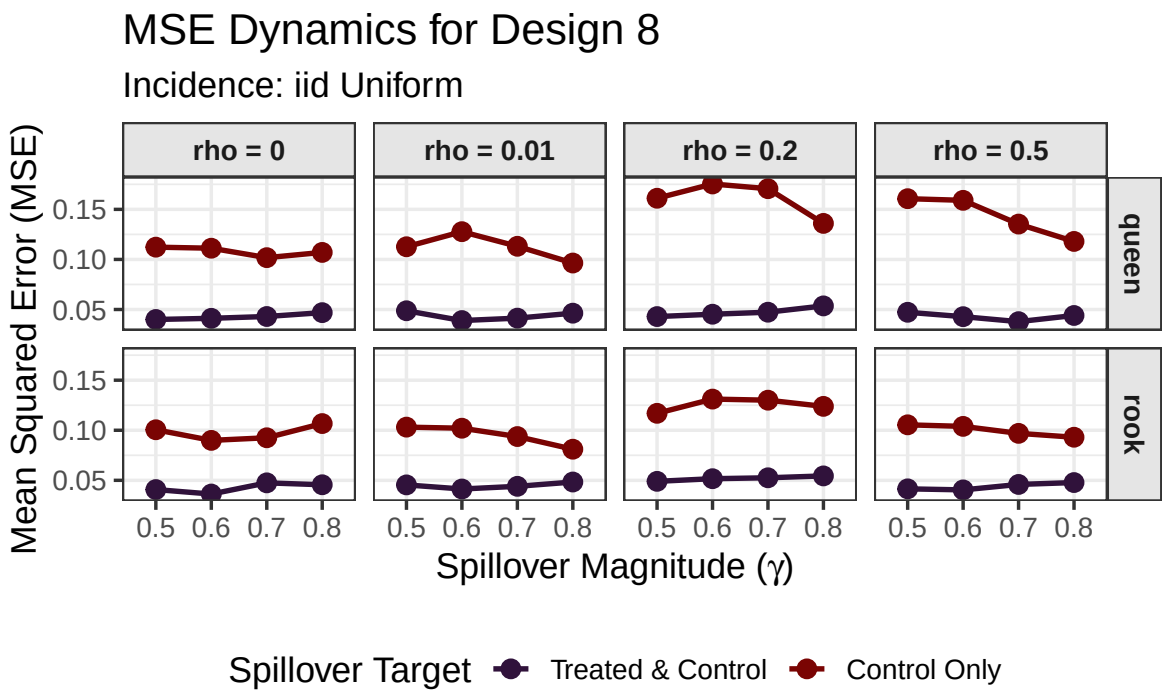

Figure 20: MLE: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

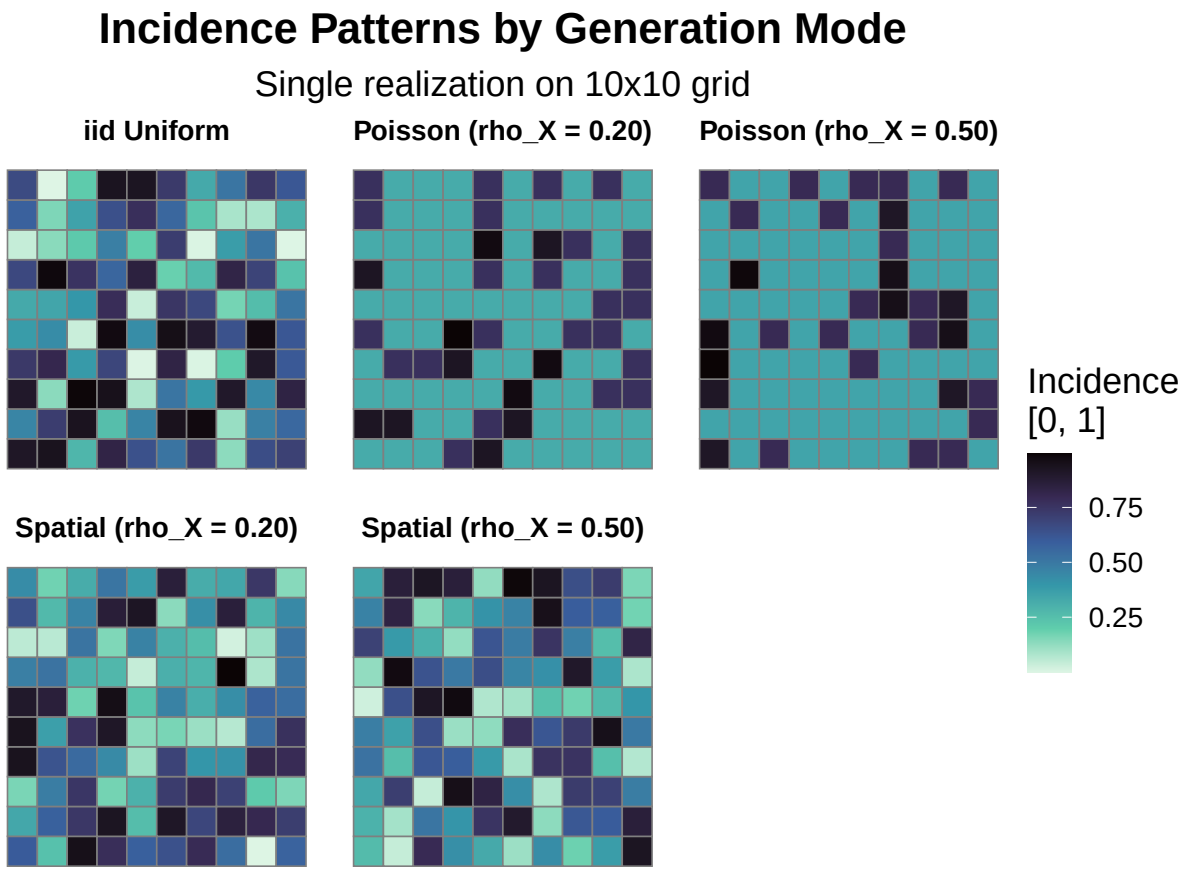

Figure 21: Incidence spatial patterns across the three generation modes for rho_X = 0.20 and 0.50. Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

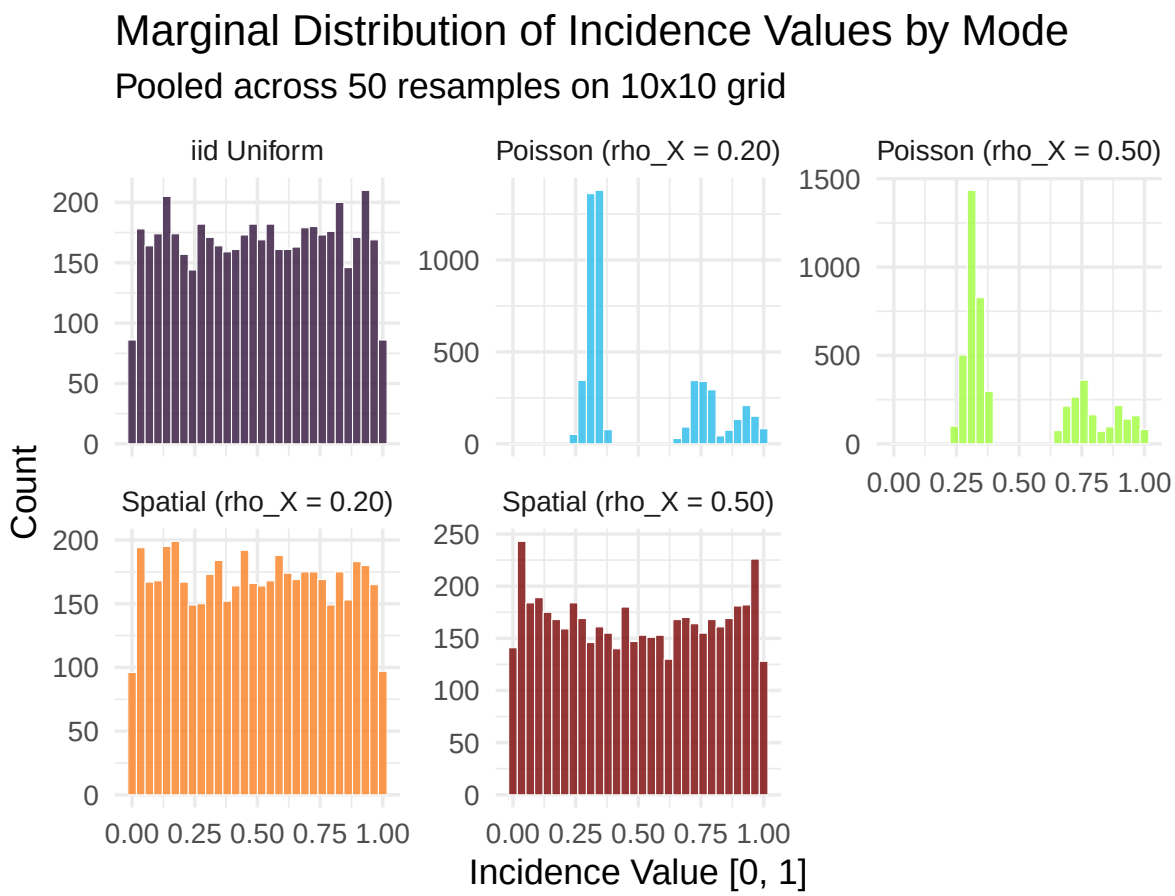

Figure 22: Marginal distribution of incidence values by mode, pooled across 50 resamples on the 10x10 grid. All modes are marginally Uniform on [0,1]; spatial structure manifests in spatial covariance, not the marginal.

6 Subgroup Results by Incidence Mode

The following tables report MLE results separately for each incidence configuration. Results are **never aggregated across incidence modes**, as each represents a fundamentally different assumption about prior data.

6.1 MLE: iid Uniform

Table 10: MLE robustness profile: iid Uniform

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0351	0.0447	0.0646	0.1105	0.1544	0.0658
Design 8	0.0363	0.0454	0.0677	0.1114	0.1753	0.0661
Design 6	0.0358	0.0421	0.0757	0.1576	0.2512	0.1155
Design 7	0.0327	0.0449	0.0827	0.1590	0.2818	0.1141
Design 2	0.0278	0.0530	0.1043	0.2042	0.3970	0.1512
Design 5	0.0426	0.0560	0.1195	0.3031	0.7737	0.2471
Design 4	0.1010	0.1281	0.1487	0.1659	0.1964	0.0377
Design 1	0.0356	0.1083	0.4729	0.9530	1.9159	0.8447

6.2 MLE: Poisson (rho_X = 0.20)

Table 11: MLE robustness profile: Poisson (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0374	0.0454	0.0608	0.1034	0.1446	0.0580
Design 3	0.0406	0.0456	0.0702	0.1065	0.1980	0.0609
Design 7	0.0362	0.0438	0.0781	0.1689	0.2610	0.1251
Design 6	0.0357	0.0427	0.0800	0.1553	0.2555	0.1126
Design 5	0.0448	0.0570	0.1260	0.3076	0.7115	0.2505
Design 4	0.0969	0.1319	0.1455	0.1657	0.2349	0.0339
Design 2	0.0235	0.0722	0.2460	0.4426	0.7982	0.3704
Design 1	0.0537	0.2567	0.5481	1.0538	4.0611	0.7970

6.3 MLE: Poisson (rho_X = 0.50)

Table 12: MLE robustness profile: Poisson (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0367	0.0442	0.0634	0.1110	0.1725	0.0668
Design 8	0.0370	0.0449	0.0644	0.1010	0.1724	0.0561
Design 6	0.0327	0.0426	0.0781	0.1451	0.2771	0.1025
Design 7	0.0298	0.0426	0.0853	0.1548	0.2443	0.1122
Design 5	0.0395	0.0533	0.1157	0.2949	0.8079	0.2416
Design 2	0.0493	0.0843	0.1287	0.2036	0.6502	0.1193
Design 4	0.1054	0.1239	0.1412	0.1559	0.1881	0.0320
Design 1	0.0202	0.1664	0.4381	0.7554	2.1564	0.5890

6.4 MLE: Spatial (rho_X = 0.20)

Table 13: MLE robustness profile: Spatial (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0345	0.0440	0.0631	0.1111	0.1618	0.0670
Design 8	0.0333	0.0451	0.0669	0.1138	0.1992	0.0687
Design 6	0.0360	0.0440	0.0768	0.1394	0.2265	0.0955
Design 2	0.0124	0.0509	0.0782	0.1351	0.2586	0.0842
Design 7	0.0344	0.0439	0.0865	0.1624	0.2496	0.1185
Design 5	0.0443	0.0539	0.1278	0.3036	0.7249	0.2497
Design 4	0.1079	0.1285	0.1445	0.1604	0.2158	0.0319
Design 1	0.0343	0.1650	0.4272	0.8489	3.0219	0.6838

6.5 MLE: Spatial (rho_X = 0.50)

Table 14: MLE robustness profile: Spatial (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0389	0.0457	0.0647	0.1037	0.1686	0.0580
Design 3	0.0359	0.0444	0.0670	0.1092	0.1724	0.0648
Design 2	0.0135	0.0499	0.0672	0.0895	0.1433	0.0396
Design 6	0.0364	0.0433	0.0727	0.1452	0.2282	0.1019
Design 7	0.0343	0.0427	0.0804	0.1670	0.2627	0.1243
Design 5	0.0460	0.0559	0.1253	0.3048	0.7440	0.2489
Design 4	0.1039	0.1314	0.1419	0.1625	0.1987	0.0311

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 1	0.0169	0.1502	0.3835	0.9498	2.2750	0.7996

7 Conclusions

7.1 Summary of Findings

The simulation study evaluated 8 treatment assignment designs across 2,560 parameter scenarios, spanning three incidence generation modes, two contiguity structures, 4 spatial autocorrelation levels, 4 spillover magnitudes, and two spillover types.

Key findings:

1. **MLE achieves near-nominal coverage.** The average coverage across all scenarios is 0.89, close to the nominal 0.95 target. This reflects MLE's explicit modeling of spatial autocorrelation ρ, which ensures valid inference when spatial dependence is present.
2. **Design 8 is the top performer.** It achieves the lowest average MSE (0.079) and a mean coverage of 0.94. Random saturation per quadrant introduces sufficient variation in spatial treatment patterns, allowing the spatial autoregressive model to cleanly separate τ from ρ.
3. **Design 1 is the consistently worst design.** It achieves the highest average MSE (0.744) and the lowest mean coverage (0.55). Its perfect spatial alternation creates systematic confounding when incidence is spatially structured, inflating bias and degrading coverage.
4. **High Incidence Focus and Balanced Quartile/Strata designs are robustly competitive.** These designs rank consistently well across incidence mode and parameter regimes. Incidence-stratified assignment directly addresses confounding from X.
5. **Design rankings are stable across incidence modes.** The ordering of designs by MSE is largely consistent across iid, spatial, and Poisson incidence configurations, suggesting the findings generalize across different assumptions about prior data structure.

7.2 Limitations and Planned Extensions

- **Oracle MLE:** The MLE model uses the true spillover covariate. Non-oracle comparisons (`include_spill_covariate = FALSE`) remain as a planned extension.
- **Iteration count:** MLE uses 250 iterations per scenario. Apparent variance inflation at high-ρ may partly reflect this count; a larger sweep would reduce Monte Carlo error.
- **Equal populations:** Poisson mode uses $P_i = 1{,}000$ for all clusters. Heterogeneous population sizes (log-normal) would more closely match NC county data.
- **Grid sensitivity:** All results use $G = 10$ ($N = 100$). Sensitivity to $G = 8$ or $G = 15$ has not been evaluated.

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8 References

- Mirzaei, A. et al. (2019). Sudden unexpected death rates. *(SUD base rate 35/100,000)*
- Gan, W. et al. (2019). County-level Poisson regression for SUD mortality.
- Watson, K. et al. (AHA). Census tract-level spatial analysis of SUD.
- LeSage, J. & Pace, R.K. (2009). *Introduction to Spatial Econometrics*. CRC Press.
- Bivand, R., et al. `spdep` R package.
- Bivand, R. & Piras, G. `spatialreg` R package.
- R packages used: `dplyr`, `tidyr`, `ggplot2`, `viridis`, `knitr`, `kableExtra`, `digest`

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9 Appendix: Result File Locations

Table 15: RDS result files in results/sim_data/

File	Location
sim_results_DIM_combined_20260304_195321.rds	sim_data
sim_results_DIM_iid_20260304_195321.rds	sim_data
sim_results_DIM_poisson_20260304_195321.rds	sim_data
sim_results_DIM_spatial_20260304_195321.rds	sim_data
sim_results_MLE_combined_20260305_150742.rds	sim_data
sim_results_MLE_combined_20260322_151030.rds	sim_data
sim_results_MLE_iid_20260305_150742.rds	sim_data
sim_results_MLE_iid_20260322_151030.rds	sim_data

File	Location
sim_results_MLE_poisson_20260305_150742.rds	sim_data
sim_results_MLE_poisson_20260322_151030.rds	sim_data
sim_results_MLE_spatial_20260305_150742.rds	sim_data
sim_results_MLE_spatial_20260322_151030.rds	sim_data

All simulation data files are located in: `results/sim_data/`